

R-1 Sounding rocket

Specifications and assembly instructions



Description

The R-1 is an dumb, unitary, low tech, single stage, liquid fuelled sounding rocket. The rocket is designed to take a payload on top to the limits of Kerbin's atmosphere.

The rocket has no flight computer and depends on command inputs from the payload. No guidance is available except for passive stabilisation provided by three small fins.

The rocket is suitable for launching small (125 cm) naked capsules like the Mk1.

Basic construction

The rocker is built in a single assembly operation and construction is straightforward. Only a single variant exists.

Assembly instructions

Parts list

- 1x LV-T30 "Reliant" Liquid Fuel Engine
- 2x FL-T200 Fuel Tank
- 1x FL-T100 Fuel Tank
- 3x Basic Fin
- 2x Mk2-R Radial-Mount Parachute
- 1x TD-12 Decoupler

Assembly procedure

- Start with Reliant engine
 - Position gas generator exhaust pointing to the west
- Connect T200 tank axially on top of engine
- Connect T200 tank axially on top of T200 tank
- Connect T100 tank axially on top of upper T200 tank
 - Use Dark ("black lines") pattern
- Connect decoupler axially on top of T100 tank
- Install fins radially on T400 tank
 - 3x symmetry with snap
 - Position W/NE/SE
 - As low as possible during fitting

- Lower fin rivet should be positioned at the bottom of the engine should "warning pattern"
- Install parachutes axially on T100 tank
 - 2x symmetry with snap
 - Position N/S
 - Raise parachutes along rocket length until centre of parachute top intersects with bottom of upper line on T100 tank
- Repaint as desired



Figure 1: Final assembled form of the R-1 rocket. Note parachute position relative to uppermost line painted in top tank.

Stage configuration

Part	Variable	Value	Default	Notes
Engine	Throttle	Main throttle	Yes	
	Thrust limiter	100	Yes	
LFOX tank (lower)	Flow priority	+2	Yes	
LFOX tank (mid)	Flow priority	+1	No	
LFOX tank (upper)	Flow priority	0	No	
Decoupler	Force percent	100	Yes	
	Enable crossfeed	False	Yes	
	Enable staging	True	Yes	

Part	Variable	Value	Default	Notes
Parachutes	Deploy mode	When safe	Yes	
	Min pressure	0.6	No	
	Altitude	500	No	
	Spread angle	10	No	

Staging

Stage	Parts	Notes
0	Parachutes, decoupler	
1	Engine	

Post flight maintenance

- Decoupler
 - Enable staging
- Tanks
 - Fill

Pre flight checks

- Verify staging
- Verify parachute settings